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Reflection on Phase Two of the Project

Embarking on Phase Two of this project was both challenging and enlightening. My initial task was to populate the products table with 1,000 types of watches. Utilizing Mockaroo, I generated realistic mock data, which provided a solid foundation for the products in the database. The next step involved inserting 100,000 records into the user_account table. This massive dataset was crucial for simulating a real-world user base. However, integrating these users into the orders and order_items tables presented unforeseen challenges. I encountered numerous trial-and-error situations, primarily because the records in the user_account table didn't align seamlessly with those in the order_items table. This misalignment highlighted the complexities of maintaining referential integrity in relational databases.

One of the significant hurdles was understanding how to insert data into tables with foreign key constraints. Initially, I was unfamiliar with the process, leading to multiple errors and frustrations. To overcome this, I turned to YouTube tutorials, which provided step-by-step guidance on handling foreign key relationships. These resources were invaluable, offering insights into the correct sequence of data insertion and the importance of maintaining data consistency across related tables.

To ensure the accuracy and integrity of the data, I uploaded the products and user_account tables as datasets into Mockaroo. This approach allowed me to generate mock data for the order_items table that adhered to the existing foreign key constraints. By referencing actual IDs from the parent tables, I ensured that the generated data maintained the necessary relationships, thereby preserving the database's integrity.

To validate the correctness of the data and the relationships between tables, I executed several SQL queries:

1. **Total Sales per Product:** This query helped identify which products were generating the most revenue.
2. **Total Orders by a Specific User:** By focusing on individual users, I could assess user engagement and purchasing behavior.
3. **Average Order Price per User:** This provided insights into the spending habits of users.
4. **Top 5 Most Ordered Products by Quantity:** Highlighting the most popular products helped in understanding market demand.
5. **Total Sales per Product Grouped by Year:** This temporal analysis shed light on sales trends over time.

Each of these queries served as a checkpoint, confirming that the data relationships were correctly established and that the mock data behaved as expected.

Reflecting on this phase, I gained a deeper appreciation for the intricacies of database management, especially concerning data integrity and the challenges of simulating real-world scenarios. The experience underscored the importance of thorough planning, continuous learning, and the effective use of available resources. While the journey was fraught with challenges, the knowledge and skills acquired have been invaluable, laying a strong foundation for future endeavors in database management and data analysis.